THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET D

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The distance of the point $(9,\ 12)$ from the origin is: 1

(a) 15 units **(b)** 21 units **(c)** 16 units **(d)** 9 units

2. The reflection of the point $(5,\ 7)$ in the y-axis is: 1

(a) $(-5,\ 7)$ **(b)** $(7,\ 5)$ **(c)** $(-5,\ -7)$ **(d)** $(5,\ -7)$

3. A point lies on the y-axis at distance 5 units below the origin. Its coordinates are: 1

(a) $(5,\ 0)$ **(b)** $(0,\ 5)$ **(c)** $(0,\ -5)$ **(d)** $(-5,\ 0)$

4. The three points $O(0,0)$, $A(3,4)$ and $B(-3,-4)$: 1

(a) form an isosceles triangle **(b)** form a right triangle
(c) are collinear **(d)** form an equilateral triangle

5. The point $(3,\ 0)$ is reflected in the x-axis. Its image is: 1

(a) $(3,\ 0)$ **(b)** $(0,\ -3)$ **(c)** $(0,\ 3)$ **(d)** $(-3,\ 0)$

6. For the line $y = x + 3$, the slope and the y-intercept are respectively: 1

(a) 1 and -3 **(b)** -1 and 3
(c) 3 and 1 **(d)** 1 and 3

7. The point $(2,\ 5)$ ___ the line $y = 3x - 2$. 1

(a) does not lie on the line **(b)** lies on the line
(c) is the y-intercept of the line **(d)** is the origin

8. The coefficient of x^2 in $x^4 - 3x^3 + 6x^2 - 2x + 7$ is: 1

(a) 7 **(b)** −2 **(c)** 6 **(d)** −3

9. A linear growth is represented by a straight line with: 1

(a) negative slope **(b)** zero intercept **(c)** positive slope **(d)** undefined slope

10. The perimeter of a square of side x cm is a linear polynomial. For side 2.5 cm it is: 1

(a) 5 cm **(b)** 6.25 cm **(c)** 12.5 cm **(d)** 10 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the coordinates of the point on the y-axis that is 6 units above the origin and of the point on the x-axis that is 4 units to the left of the origin. Find the distance between them. 2

12. Check whether $P(1,\ 0)$ is equidistant from $A(2,\ 3)$ and $B(4,\ -1)$. 2

13. An auto-rickshaw fare is ₹25 for the first 2 km and ₹15 for every further km. Find the fare for 10 km. 2

14. The sum of two numbers is 64 and one is 10 more than the other. Form a linear equation and find the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Given $A(2,\ -2)$ and $B(8,\ 4)$, find the point P on AB that divides it in the ratio $1 : 2$ (nearer to A) and verify that $AP = \frac{1}{3}AB$. 3

16. The wire of length 20 cm is bent into a rectangle of length x cm. Write the area in terms of x and find it for $x = 6$. 3

17. Find the length of the hypotenuse and the area of the triangle with vertices $O(0, 0)$, $A(6, 0)$ and $B(0, 8)$. 3

OR

Show that $A(3,\ 0)$, $B(6,\ 4)$ and $C(-1,\ 3)$ are the vertices of an isosceles right-angled triangle. 3

18. Given $y = 2x + 1$, complete the table: $x = 1, 2, 5, 7, 9, 12, 20$ and check which of the points lies on the line: $(4, 9)$ or $(6, 14)$. 3

OR

Solve the linear equation $3x - 4 = 11$ and verify the solution. Also write a word problem that leads to $2x + 10 = 64$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A rectangular garden $OABC$ has $O(0, 0)$, $A(12, 0)$, $B(12, 8)$, $C(0, 8)$. A fountain is at the centre F, and lamps $L_1(0, 4)$, $L_2(12, 4)$. 5

(i) Find the centre.

(ii) Show $FL_1 = FL_2$.

(iii) Find the area of the garden and the distance between O and B.

OR

In $\triangle ABC$, $A = (1,\ 2)$ and $B = (4,\ 6)$. Find the midpoint M of AB and the length of the median CM, where $C = (7,\ 2)$. 5

20. Find the equation of the line passing through $(1,\ 3)$ and $(3,\ 7)$. Also 5

(i) check whether $(5,\ 11)$ lies on it,

(ii) find its y-intercept,

(iii) write the equation of a parallel line through $(0,\ -1)$.

OR

A rectangular garden has length l m and width w m. Wire fencing costs ₹100/m along the length (both sides), wooden fencing ₹80/m along the width (both sides) and seeding ₹50/m^2. 5

(i) Write the expression for the total cost.

(ii) Find it for $l = 20$, $w = 10$.

(iii) Name the terms, variables and coefficients.

END OF QUESTION PAPER — **Total: 40 Marks**